

Investigating the Impact of Grafted Molecule Types on Frequency Domain Spectroscopy of Polypropylene Insulation

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Abstract—Thermoplastic insulation materials represented by polypropylene (PP) have become an important direction for future research on cable insulation materials. To develop high performance PP cable insulation materials, the Havriliak-Negami (H-N) model is used to study the impact of grafted molecule types on frequency domain spectroscopy (FDS) of PP insulation in this paper. Research results demonstrate that PP grafted with the irganox 3052 effectively scavenges free radicals within the material, and increases its oxidation induction time (OIT), thus enhancing the thermal stability of PP. Both the voltage stabilizer DA and the irganox 3052 can increase the barrier height (E_b) for the thermionic migration in PP. PP-g-3052 exhibits a larger E_b compared to PP-g-DA, enabling better suppression of thermionic migration and consequently reducing the thermionic polarization. Concurrently, the relationship between conductivity activation energy (E_σ) and E_b is further studied, with consistent changes observed in both. PP-g-3052 possesses the largest E_σ , exerts the strongest inhibitory effect on carrier migration, and more effectively enhances the insulation properties of PP.

Keywords—polypropylene, graft modification, frequency domain spectroscopy, H-N model

I. INTRODUCTION

As high-voltage direct current (HVDC) transmission systems continuously develop in the electric power sector, the demand for cable insulation materials is increasing. While polypropylene (PP) insulation materials have some physical and chemical properties, their performance is still insufficient under the harsh environmental conditions of HVDC systems, including high voltage and high electric field stress [1]

To enhance the electrical performance of PP cable insulation, researchers have employed methods such as graft modification and nano-modification [2]. Graft-modified PP has gained significant attention as insulation material for HVDC cables among these methods. Graft modification of PP involves introducing different graft monomers onto the PP chain, such as maleic anhydride [3] and styrene [4], to enhance its performance. These graft monomers undergo chemical reactions with the PP chain, forming copolymers or graft chains, thereby altering the physical and chemical properties of PP. Studies have indicated that the incorporation of graft monomers such as maleic anhydride and styrene into PP leads to an increase in trap depth and density within the material. This process inhibits carrier migration, thereby improving the insulation properties of PP [3-4]. Additionally, graft modification has the capacity to impede small molecule migration and enhance the stability of PP, thereby enabling the material to maintain optimal performance over extended

periods. However, various types of grafted molecules exert diverse effects on PP insulation. Thus, investigating the impact of grafted molecule types on PP insulation is significant.

In the case of polymer insulation materials, frequency domain spectroscopy (FDS) reveals the dielectric response and polarization behavior of materials at different frequencies or temperatures. Through FDS analysis, the polarization mechanism, charge transfer process, and interface effects of materials can be understood [2]. In this paper, the irganox 3052 and the voltage stabilizer DA are used as grafting monomers and the effect of the grafted molecule types on the FDS of PP insulation is studied using the Havriliak-Negami (H-N) model. Simultaneously, the effect of the type of grafted molecule on the physicochemical properties of polypropylene was explained by melt flow rate (MFR) and oxidation induction time (OIT) tests.

II. EXPERIMENTAL SETUP

A. Sample Preparation

The study employs the voltage stabilizer Dibenzylidene acetone (referred to as DA) and the hindered phenolic irganox 3052 (referred to as 3052) as grafting monomers. The molecular structures of DA and 3052 are depicted in Fig. 1. The monomers react with isotactic polypropylene (PP) at 180°C for 15 min using dicumyl peroxide (DCP) as the grafting initiator. Subsequently, the grafted sample undergoes further drying in a vacuum oven to eliminate any residual monomers and surface moisture. A small quantity of the dried graft sample is then weighed, heated, and dissolved in xylene, with subsequent reflux until complete dissolution. The solution is then poured into an unheated acetone solution, left to precipitate, filtered under suction, and finally dried in a vacuum oven to yield the purified graft sample. The purified sample is hot-pressed using a flat vulcanizer at 200 °C and 15 MPa for 10 min. It is then cooled in a water bath at 20 °C and 10 MPa for 10 min to obtain sheet samples with a thickness of 200 μm .

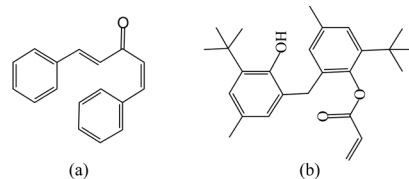


Fig. 1 Grafted molecular structure. (a) DA. (b) 3052.

B. Experimental Methods

The XNR-400C melt flow indexer is used to measure the MFR of each sample. When the machine reaches a

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temperature of 230°C, the prepared sample powder is loaded into the barrel, and a load of 2.16 kg is applied. Following this, the mass of the molten material flowing through a circular tube with a diameter of 2.1 mm within 10 min is measured. Each sample undergoes five tests, and the average measurement result is recorded.

The OIT is conducted using the differential scanning calorimeter. Initially, a sample weighing 5 mg is heated to 210 °C in a nitrogen atmosphere and held at a constant temperature for 5 min. The sample is then exposed to an oxygen atmosphere at 50 mL/min gas flow, maintaining a constant temperature of 210°C until an exothermic peak appears in the heat flow curve.

The relative permittivity and dielectric loss characteristics of each sample are determined using a wideband dielectric impedance spectrometer (Novocontrol Concept 40). The surface of each of the samples is sprayed with gold and exposed to an electric field strength of 1 kV/mm.

C. Havriliak-Negami Model

The H-N model can be used to explain phenomena such as dielectric relaxation and mechanical relaxation in polymer systems. The model is more widely in use and has better properties than the Debye model. Both conductivity and polarization phenomena are included in the dielectric spectrum of a polymer system. Taking into account the conductivity, the equation of the modified H-N model can be formulated as follows [2]:

$$\varepsilon^*(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon}{(1 + (i\omega\tau)^\beta)^\gamma} + \frac{\sigma}{i\varepsilon_0\omega} \quad (1)$$

where $\Delta\varepsilon$ represents the relaxation strength, τ stands for the relaxation time, β and γ denote the shape coefficients, σ represents the conductivity, and ε_0 denotes the vacuum dielectric constant.

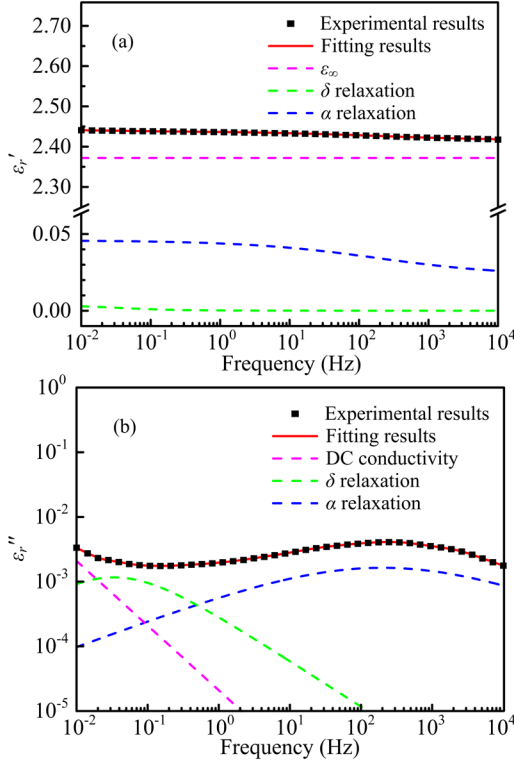


Fig. 2 FDS fitting results using the H-N model. (a) Real permittivity. (b) Imaginary permittivity.

Here, the FDS for each sample is fit using the H-N equation, and the fit included the orientational polarization process (α relaxation), the thermionic polarization process (δ relaxation), and the conductivity component. The fitting equation can be formulated as follows [2]:

$$\varepsilon^*(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon_\alpha}{(1 + (i\omega\tau_\alpha)^\beta)^\gamma} + \frac{\Delta\varepsilon_\delta}{(1 + (i\omega\tau_\delta)^\beta)^\gamma} + \frac{\sigma}{i\varepsilon_0\omega} \quad (2)$$

Fig. 2 illustrates the results of the fitting of the complex dielectric constant of PP-g-3052 at 10 °C by means of the H-N model.

III. RESULTS AND DISCUSSION

A. Sample Characteristics

The MFR serves as a tool for assessing the molecular weight of polymer materials and evaluating the extent of PP degradation under high-temperature processing conditions. An increase in degradation reactions within the PP main chain leads to a higher MFR for the material. Fig. 3 illustrates the MFR values for each of the samples, which are 2.23 g/10 min for PP, 2.32 g/10 min for PP-g-DA, and 2.27 g/10 min for PP-g-3052. In comparison to pure PP, both PP-g-DA and PP-g-3052 exhibit slightly elevated MFR values, indicating a certain level of PP molecular chain degradation during the grafting process. However, the level of degradation is minor, and PP-g-DA and PP-g-3052 still maintain good performance.

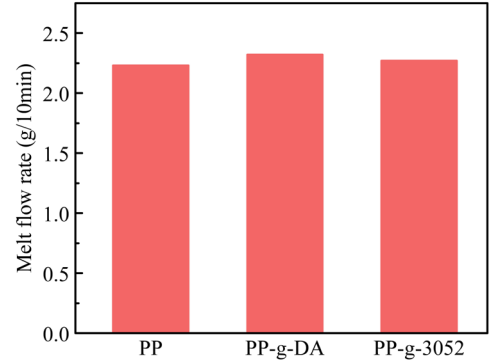


Fig. 3 MFR values of each sample.

OIT characterizes the thermal stability of a material and evaluates its resistance to heat-induced deterioration. Fig. 4 presents the results of the OIT for each sample. In contrast to the OIT of 19.6 min observed for PP, PP-g-DA demonstrated an OIT of 3.1 min, while PP-g-3052 exhibited an OIT of 31.5 min, representing a 60.7% increase in OIT compared to that of PP. These results suggest that PP-g-3052 may effectively

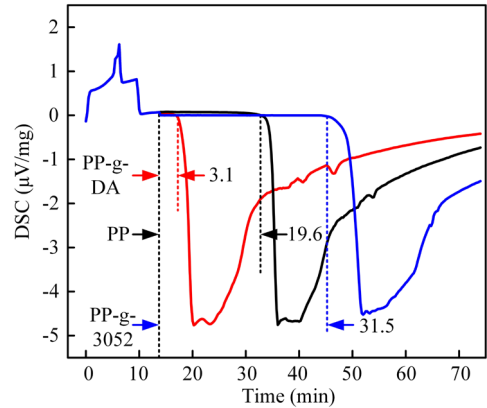


Fig. 4 OIT of each sample.

eliminate free radicals from the material, thereby enhancing the thermal stability of PP. Being a potent hydrogen donor and belonging to the hindered phenolic organox family, 3052 can halt the chain reaction triggered by free radical autoxidation, a process that occurs as macromolecules age. In this process, impeded phenol generates aryl oxygen radicals and can also trap free radicals, effectively stopping the oxidation chain reaction. Fig. 5 illustrates how 3052 scavenges free radicals.

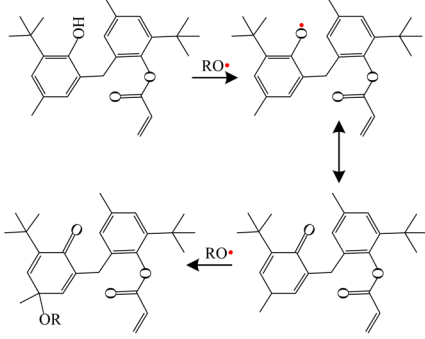


Fig. 5 Free radical scavenging mechanism of 3052.

B. FDS Characteristics

The relative permittivity ϵ_r' and dielectric loss ϵ_r'' curves of each sample are presented in Fig. 6. Compared to PP, the relative permittivity of PP-g-DA and PP-g-3052 are slightly higher. The high frequency dielectric loss of PP-g-DA and PP-g-3052 is greater than that of PP due to the orientational polarization introduced by the PP graft with DA and 3052. Significant peaks of orientational polarization loss are observed for PP-g-DA and PP-g-3052 at 10 and 30 °C. In the ultra-low frequency range, the dielectric loss of PP-g-DA and PP-g-3052 is lower than that of PP. This improvement can be attributed to graft modification, which improves the insulating properties of the materials, thereby reducing both the loss of direct current conductivity and the loss of polarization.

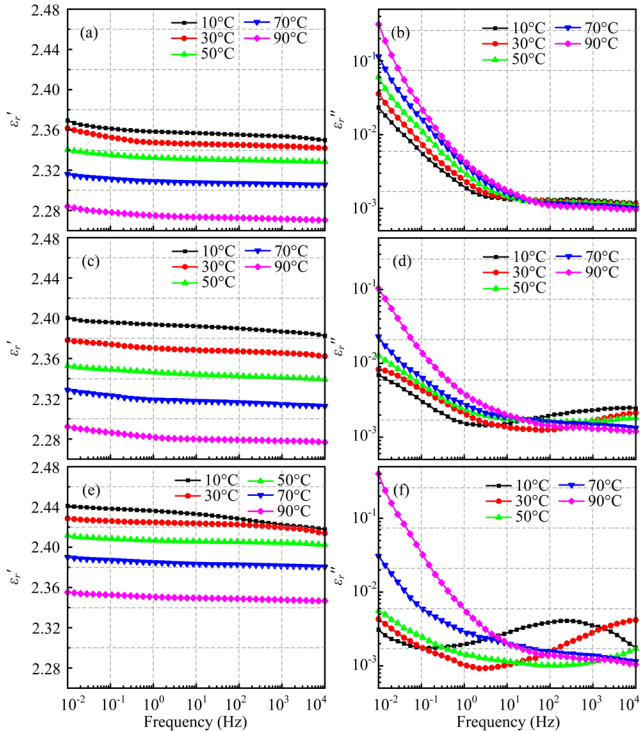


Fig. 6 FDS of each sample varying with temperature. (a) (b) PP. (c) (d) PP-g-DA. (e) (f) PP-g-3052.

C. FDS Fitting Results

As shown in Fig. 7, the α relaxation is fitted. From Fig. 7(a), it is clear that the relaxation strength $\Delta\epsilon_\alpha$ of both PP-g-DA and PP-g-3052 surpasses that of PP at the same temperature due to the polarity of both DA and 3052 molecules. With increasing temperature, orientational polarization is suppressed, leading to a decrease in the $\Delta\epsilon_\alpha$ of each sample. Fig. 7(b) shows that the relaxation time τ_α of both PP-g-DA and PP-g-3052 exceeded that of PP at the same temperature, with PP-g-3052 having the longest relaxation time.

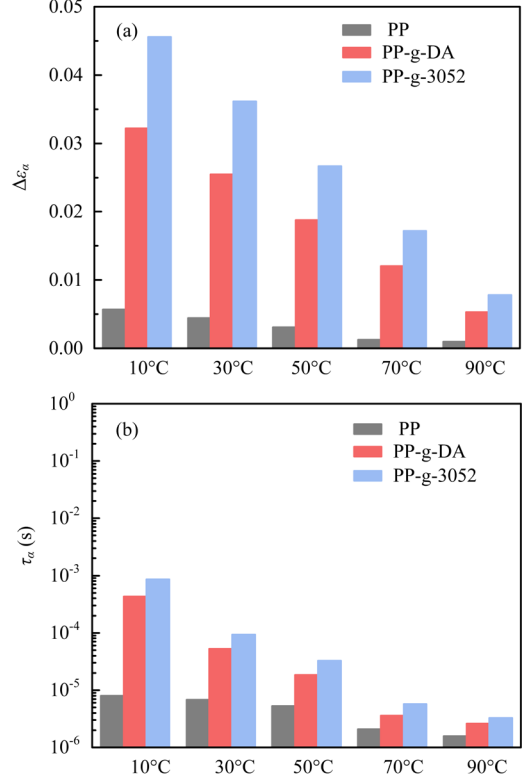


Fig. 7 The fitting results of α relaxation for each sample.

As shown in Fig. 8, the δ relaxation is fitted. The data from Fig. 8(a) demonstrate that due to the inhibition of thermionic polarization by DA and 3052 molecules, the relaxation strength $\Delta\epsilon_\delta$ of both PP-g-DA and PP-g-3052 is smaller than that of PP, and the relaxation time τ_δ is larger than that of PP at the same temperature. The charge carrier must cross a barrier for δ relaxation to occur. The δ relaxation is related to the barrier height as follows [2]:

$$\tau_\delta = \tau_0 e^{E_\delta/(kT)} \quad (4)$$

where τ_0 represents the inherent relaxation time, E_δ denotes the barrier height, and k is the Boltzmann constant.

The height of the potential barriers for each sample is determined by fitting τ_δ using (4), and the corresponding results are presented in Fig. 9. The analysis shows that the E_δ of PP is 0.3643 eV. Compared with PP, the E_δ of PP-g-DA and PP-g-3052 reach 0.4137 eV and 0.5638 eV, respectively. The grafting of DA and 3052 increases the potential barrier height of PP, which hinders the transport of thermal ions. Particularly, the E_δ of PP-g-3052 is the largest, and the inhibition of the thermionic migration is more effective. The electrical properties of PP are effectively improved by the grafting of 3052.

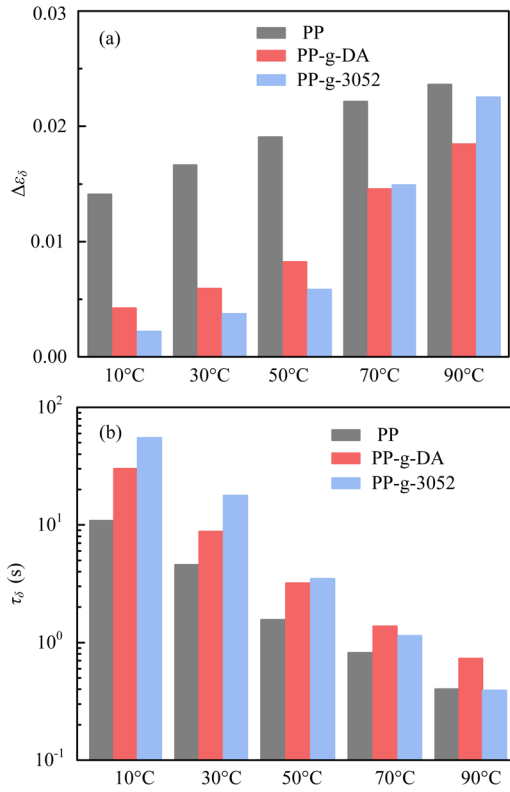


Fig. 8 The fitting results of δ relaxation for each sample.

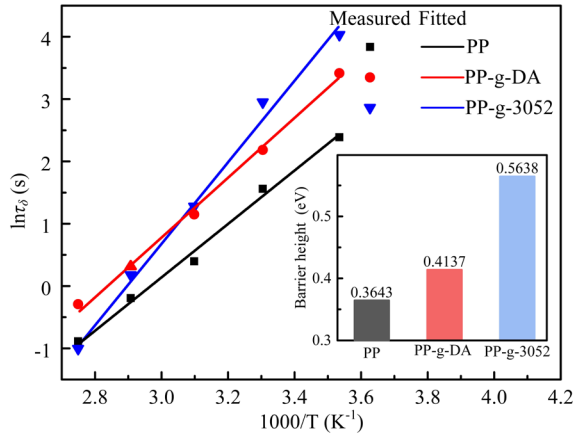


Fig. 9 Relationship between τ_δ and temperature of each sample.

Moreover, there is a strong correlation between E_δ and the activation energy that is linked to the DC conductivity process. Both mechanisms involve the movement of carriers across a potential energy barrier. The σ of each sample is analyzed using (5) to further investigate this relationship [2].

$$\sigma = Ae^{(-E_\sigma/kT)} \quad (5)$$

where A represents a temperature-independent constant, E_σ denotes the DC conductivity activation energy.

Fig. 10 displays the fitting results of the DC conductivity process. The results revealed that the E_σ of PP is 0.2548 eV. Compared with PP, the E_σ of PP-g-3052 and PP-g-DA reach 0.5808 eV and 0.3017 eV, respectively, in which PP-g-3052 has the largest E_σ . This is consistent with the change of E_δ in δ relaxation. The change in E_σ indicates that both DA and 3052 introduce more traps in PP, increasing the depth and density of traps in the material. This results in a larger potential barrier for carrier de-trapping, necessitating a larger E_σ . Moreover, the larger E_σ of PP-g-3052 compared to PP-g-

DA implies that PP-g-3052 more effectively impedes carrier migration, thereby enhancing the insulating properties of PP.

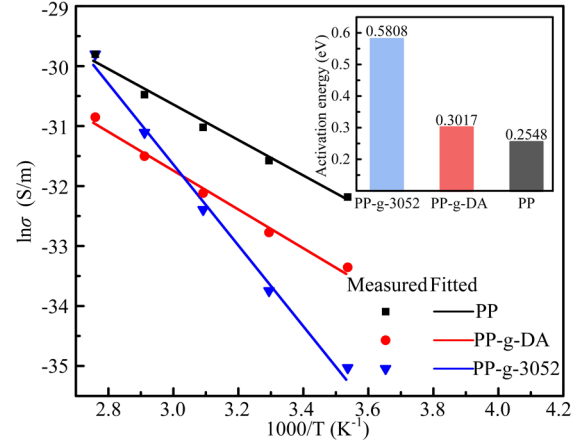


Fig. 10 Relationship between σ and temperature of each sample.

IV. CONCLUSIONS

In this paper, the influence of the types of grafted molecules on the FDS of PP insulation is studied on the basis of the H-N model. The primary findings are summarized below:

a) In comparison to the voltage stabilizer DA, PP grafted with irganox 3052 can effectively scavenge free radicals in the material, thus extending the OIT and consequently the thermal stability of the material.

b) The voltage stabilizer DA and the irganox 3052 both increase the barrier height for the thermionic migration, which effectively inhibits the thermionic polarization. PP-g-3052 has the largest barrier height and is more effective in inhibiting thermionic transport.

c) The conductive activation energy of PP-g-DA and PP-g-3052 has risen. PP-g-3052 has the highest DC conductivity activation energy, which can effectively restrict carrier migration and enhance the insulating properties of PP.

REFERENCES

- [1] Z. L. Li, Z. Zheng, Y. Wu, and B. X. Du, "Space Charge and Electric Field Dependent on Polarity Reversal of HVDC Cable Insulation," *IEEE T. Dielect. El. In.*, vol. 31, no. 1, pp. 58–65, February 2024.
- [2] Z. L. Li, Z. Y. Zhong, and B. X. Du "Dielectric relaxation and trap-modulated DC breakdown of polypropylene blend insulation," *Polymer.*, vol. 185, pp. 121935, December 2019.
- [3] Y. Zhou, J. Hu, B. Dang, and J. L. He, "Mechanism of highly improved electrical properties in polypropylene by chemical modification of grafting maleic anhydride," *J. Phys. D: Appl. Phys.*, vol. 49, no. 41, pp. 415301, September 2016.
- [4] S. X. Hu; W. J. Zhang, and W. Wang, et al., "Comprehensive Comparisons of Grafting-Modified Different Polypropylene as HVDC Cable Insulation Material," *IEEE T. Dielect. El. In.*, vol. 29, no. 5, pp. 1865–1872, October 2022.